PINION GEAR/RING GEAR (Continued)

(10) Install yoke washer and a **new** nut on the pinion gear and tighten the pinion nut until there is zero bearing end-play.

(11) Tighten the nut to 271 N·m (200 ft. lbs.).

CAUTION: Never loosen pinion gear nut to decrease pinion gear bearing rotating torque and never exceed specified preload torque. Failure to heed caution may result in damage.

(12) Using Spanner Wrench 6958 and a torque wrench set at 475 N·m (350 ft. lbs.), (Fig. 75) slowly tighten the nut in 6.8 N·m (5 ft. lbs.) increments until the rotating torque is achieved. Measure the rotating torque frequently to avoid over crushing the collapsible spacer (Fig. 76).

CAUTION: If more than 475 N·m (350 ft. lbs.) torque is required to crush the collapsible spacer, the spacer is defective and must be replaced. Failure to heed caution may result in damage.

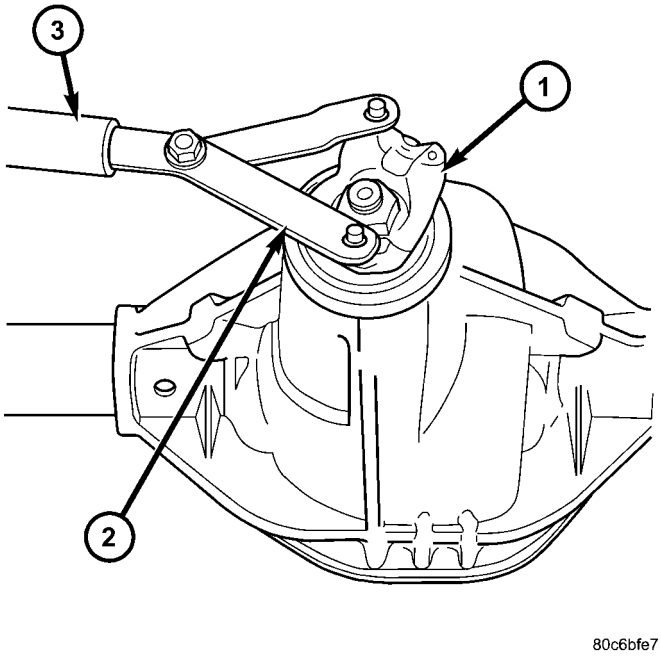

Fig. 75 PINION YOKE WRENCH

1 - PINION YOKE
2 - SPANNER WRENCH
3 - PIPE

(13) Check bearing rotating torque with a inch pound torque wrench (Fig. 76). The pinion gear rotating torque should be:

- Original Bearings: 1 to 2 N·m (10 to 20 in. lbs.).
- New Bearings: 2 to 4 N·m (20 to 35 in. lbs.).

(14) Invert the differential case and start two ring gear bolts. This will provide case-to-ring gear bolt hole alignment.

(15) Invert the differential case in the vise.

(16) Install new ring gear bolts and alternately tighten to 136 N·m (100 ft. lbs.) (Fig. 77).

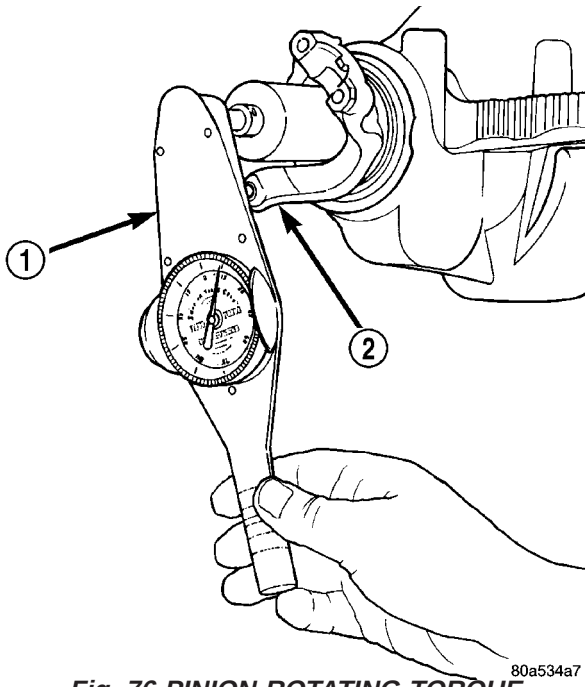

Fig. 76 PINION ROTATING TORQUE

1 - TORQUE WRENCH
2 - PINION YOKE

CAUTION: Do not reuse ring gear bolts, the bolts can fracture causing extensive damage. Failure to heed caution may result in damage.

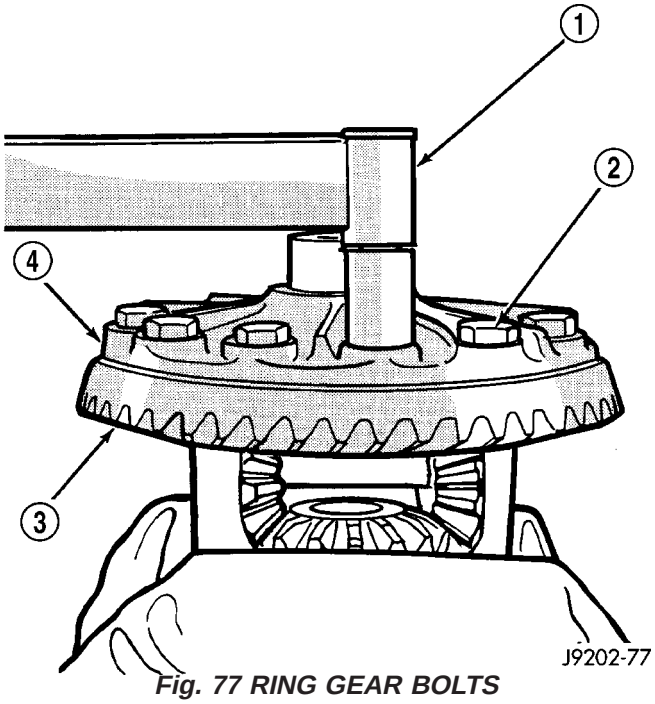

Fig. 77 RING GEAR BOLTS

1 - TORQUE WRENCH
2 - RING GEAR BOLT
3 - RING GEAR
4 - CASE

(17) Install differential in axle housing and verify gear mesh and contact pattern. Refer to Adjustments (Gear Contact Pattern) for procedure.